

XBee INTERFACING With Raspberry Pi Model 2

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XBee Pro (S2) module is an XBee ZB ZigBee Mesh Protocol based wireless transceiver designed by Digi International with an excellent range of 1.6 kilometres (one mile) line of sight (for 50mW, U.FL connection) and 90 metres indoors. This device works on 3.3V DC @ 295mA in transmission mode and has a maximum data rate of 250kbps. The data from the processor to XBee module is transmitted via a serial port.

Raspberry Pi Model 2 is a second-generation Raspberry Pi that was released in February 2015. It is a Linux based computer and has a powerful Broadcom BCM2836 ARMv7 900MHz quad-core processor and 1GB SDRAM. It has a total of 40 pins, including 17 pins GPIO along with HDMI, micro SD card slot, Ethernet, USB 2.0 and 3.5 audio (out) ports.

Software

There are many operating systems designed for Raspberry Pi Model 2, but most widely used are Raspbian (Linux based) and Microsoft Windows 10 IoT Core. If you are a newbie to Raspberry Pi Model 2, check out the link <https://www.raspberrypi.org/help> for information

on Raspberry Pi. In this article, the compiler used is Python and OS required on Raspberry Pi is Raspbian.

The objective is to send a character on XBee module and receive it on another XBee module. To send/receive data on XBee, the modules should be configured, and to do that the software required is XCTU. XCTU is a free software designed by

Digi International that allows a user to configure XBee modules. XCTU can be downloaded from the link <http://www.digi.com/products/xbee-rf-solutions/xctu-software/xctu>

Circuit and working

The components required are Raspberry Pi Model 2, two XBee modules with Explorer shields and some

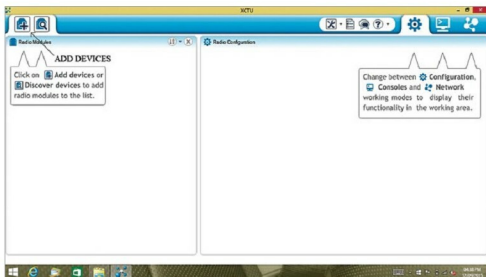


Fig. 2: XCTU software screenshot

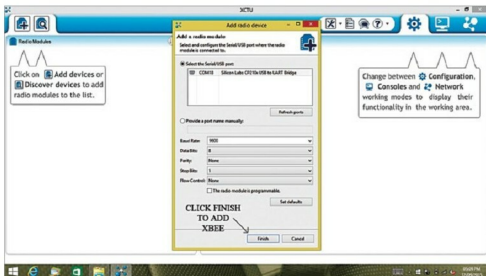


Fig. 3: Configuring the COM port



Fig. 1: XBee Explorer shield